This publication is available free of charge from: https://doi.org/10.6028/NIST.TN.2106

Comparing Instruments

Often, experimenters wish to compare the performances of two or more instruments. For example: what is the difference between how well two detectors perform, or is there any significant differences in the instrument response between a group of several instruments?

1 Statistical Preliminaries

This document will begin with brief primers about hypothesis tests and confidence intervals, as those tools will be used repeatedly in this document. There will be an additional primer about using the normal approximation to create a *z* test, which is a type of hypothesis test used repeatedly in this document.

1.1 Hypothesis tests

In this document we describe several hypothesis tests for comparing instrument performance. NIST Technical Note 2045 [1], the NIST/SEMATECH e-Handbook *[2],* and Mendenhall and Sincich [3] all give an introduction to hypothesis testing. However, since the focus of this document is the comparison of instrument performance rather than, say, confirming performance thresholds, the terminology of the hypothesis tests will be geared toward that application. In this document, the null hypothesis will usually have the form:

H_0: All instruments perform equally well.

The alternative hypothesis will have the form:

H_A: Not all instruments perform equally well.

What is called in the statistical literature a *type I error* occurs when the hypothesis test result leads us to conclude that instruments perform differently when they are not actually different. Hypothesis tests can be specified so that the probability of a *type I error* is bounded above by a number α. Such a hypothesis is said to have *significance level* equal to α. The most commonly used value of α is 0.05, although 0.0, 0.10, and 0.20 values are also used depending on the type of instrument that is being tested and its use. These significance levels are used to weigh the risk associated with a *type I error*.

For our hypothesis tests, what we call a *type II error* occurs when there really are differences between the instruments, but the hypothesis test fails to reject the null hypothesis. The probability of such an error is usually denoted by β. The quantity $1-\beta$ is called the *power* of a test, which is the probability of rejecting the null hypothesis when there is an actual difference. The *power* of a hypothesis test is not a single number but depends on the magnitude of such an actual difference; a large difference between instrument will be much easier to detect than a small difference.

The power function of a hypothesis test is the power of the test as a function of the underlying difference.